Sri Lanka Institute of Information Technology



Artificial Intelligence and Machine Learning - IT2011

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Concrete Compressive Strength Using Machine Learning.

Final Report

Group 2026-Y2-S1-MLB-B12G1-01

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1. Introduction And Problem Statement

1.1 Introduction

Concrete compressive strength is one of the most important indicators used to evaluate the quality and performance of concrete. It is influenced by several factors, including the quantities of cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and the curing age of the concrete. In this project, a machine learning-based approach is developed to predict concrete compressive strength using these eight input variables. The aim is to use historical concrete mixture data to build regression models that can estimate compressive strength in MPa and support faster analysis of concrete mix performance.

1.2 Problem Statement

The fundamental problem with traditional concrete compressive strength assessment is that it relies on laboratory testing of cured concrete specimens. Standard strength testing commonly requires a curing period of up to 28 days before the compressive strength can be measured, making the process time-consuming and potentially delaying construction-related decisions. Proposal Presentation This approach has several major limitations:

- **Long Testing Period:** Concrete specimens must be cured before strength testing can be completed. Waiting for the standard curing period delays the availability of strength information needed for further planning. Proposal Presentation
- **Delayed Construction Planning:** Since strength results are not immediately available, engineers and project teams may have to wait before making decisions related to concrete performance and subsequent construction activities. Proposal Presentation
- **Higher Testing Time and Cost:** Preparing specimens, allowing them to cure, and performing laboratory crushing tests require additional time and resources, which can contribute to increased project costs. Proposal Presentation
- **Lack of Early Predictive Capability:** Traditional testing provides strength information only after physical testing is carried out. It does not provide an early estimate of compressive strength directly from the concrete mixture composition and curing age. Therefore, this project investigates machine learning regression models that can estimate compressive strength from historical concrete data as a decision-support tool.

1.3 Project Objectives and Scope

This project aims to develop a machine learning-based system to predict the compressive strength of concrete using its material composition and curing age. The system focuses on:

- Predicting concrete compressive strength in MPa using eight input features.
- Preprocessing and analyzing concrete mixture data for reliable model training and comparing multiple regression algorithms
- Evaluating model performance using MAE, RMSE, and R^2 Score.

2. Dataset Description

2.1 Dataset Source

The dataset used in this project is the Concrete Compressive Strength Dataset. It was obtained from Kaggle and is based on the original research conducted by I Cheng Yeh (1998). The dataset contains experimental concrete mixture data that can be used to study the relationship between concrete ingredients, curing age, and compressive strength.

2.2 Dataset Structure

The dataset contains 1,030 concrete mixture samples. Each sample consists of eight input features and one target variable, giving a total of nine variables. The input features describe the material composition of the concrete and its curing age, while the target represents the measured concrete compressive strength.

Feature	Unit	Description
Cement	**kg/m³**	**Main binding material**
Blast Furnace Slag	**kg/m³**	**Cement replacement material**
Fly Ash	**kg/m³**	**Supplementary cementitious material**
Water	**kg/m³**	**Water used in the concrete mix**
Superplasticizer	**kg/m³**	**Improves workability**
Coarse Aggregate	**kg/m³**	**Large aggregate particles**
Fine Aggregate	**kg/m³**	**Sand component**
Age	**Days**	**Concrete curing period**
Compressive Strength (output)	**MPa**	**Target variable to be predicted**

2.3 Dataset Characteristics

The dataset is well suited for a regression-based machine learning problem because the target variable, concrete compressive strength, is a continuous numerical value. The dataset consists of real laboratory measurements rather than synthetic data, contains no missing values in the provided version, and includes the major material and curing factors related to concrete strength. Its size of 1,030 observations is also suitable for developing and comparing multiple machine learning regression models. Proposal Presentation

However, the dataset also has several limitations. The measurements were collected under laboratory conditions and may not fully represent real construction-site environments. Environmental variables such as temperature and humidity were not recorded, and the dataset is limited to the material compositions included in the available samples. These limitations should therefore be considered when interpreting model predictions and generalizing them to real-world applications.

3. PREPROCESSING AND EDA

Before training the machine learning models, the Concrete Compressive Strength dataset was explored and preprocessed to improve data quality and understand the relationships between the input variables and the target variable. The dataset contains numerical measurements of concrete ingredients, curing age, and compressive strength, making it suitable for statistical analysis and regression modelling.

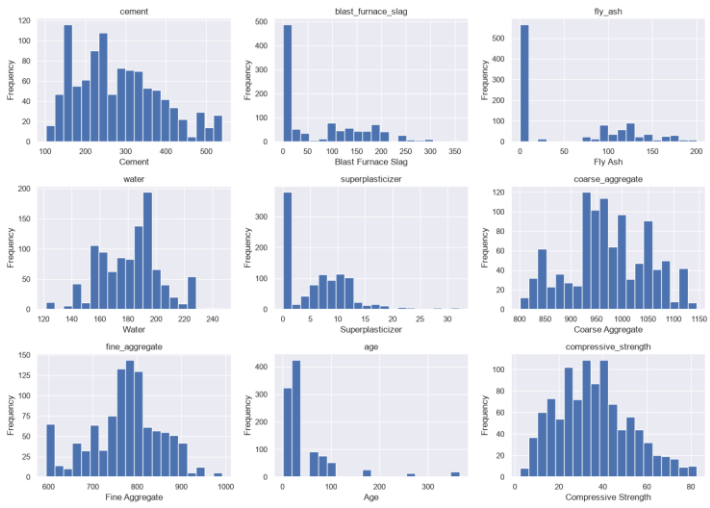
3.1 Exploratory Data Analysis

Exploratory Data Analysis was first performed to understand the structure, distribution, and quality of the dataset. The following steps were carried out:

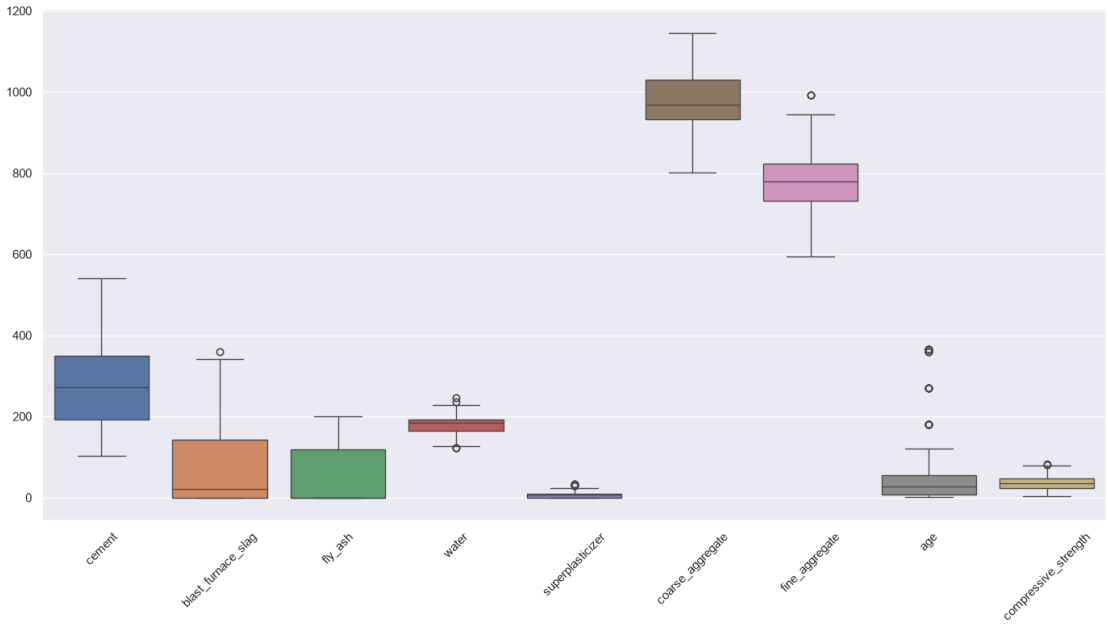
- **Dataset Inspection:** The number of rows and columns, data types, non-null counts, and the first and last records were examined.
- **Missing Value Analysis:** Each feature was checked for missing values. The provided dataset was identified as having no missing values.
- **Duplicate Detection:** Duplicate observations were identified before preprocessing so that repeated records would not unnecessarily influence model training.
- **Summary Statistics:** Descriptive statistics such as mean, standard deviation, minimum, maximum, and quartile values were examined for all numerical variables.

	count	mean	std	min	25%	50%	75%	max
cement	1030.0	281.167864	104.506364	102.00	192.375	272.900	350.000	540.0
blast_furnace_slag	1030.0	73.895825	86.279342	0.00	0.000	22.000	142.950	359.4
fly_ash	1030.0	54.188350	63.997004	0.00	0.000	0.000	118.300	200.1
water	1030.0	181.567282	21.354219	121.80	164.900	185.000	192.000	247.0
superplasticizer	1030.0	6.204660	5.973841	0.00	0.000	6.400	10.200	32.2
coarse_aggregate	1030.0	972.918932	77.753954	801.00	932.000	968.000	1029.400	1145.0
fine_aggregate	1030.0	773.580485	80.175980	594.00	730.950	779.500	824.000	992.6
age	1030.0	45.662136	63.169912	1.00	7.000	28.000	56.000	365.0
compressive_strength	1030.0	35.817961	16.705742	2.33	23.710	34.445	46.135	82.6

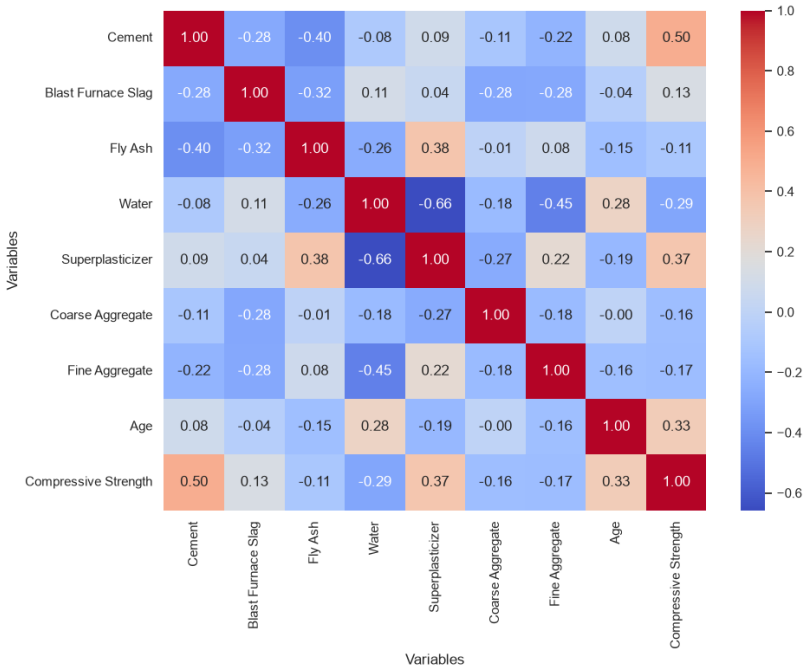
- **Distribution Analysis:** Histograms were generated to study the distributions of the concrete mixture variables and compressive strength.



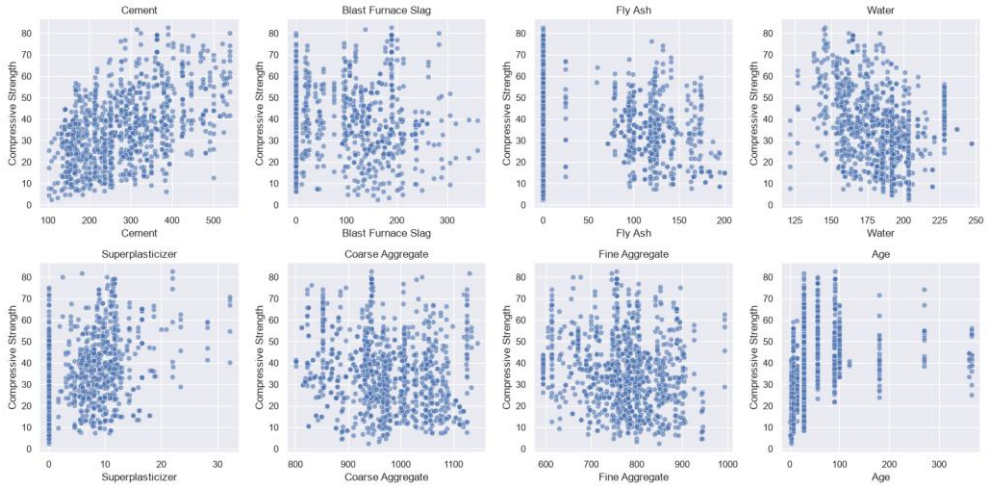
- **Outlier Analysis:** Boxplots and the Interquartile Range (IQR) method were used to identify unusually high or low observations.



- **Correlation Analysis:** A correlation heatmap was generated to examine relationships among the numerical features and the target variable.



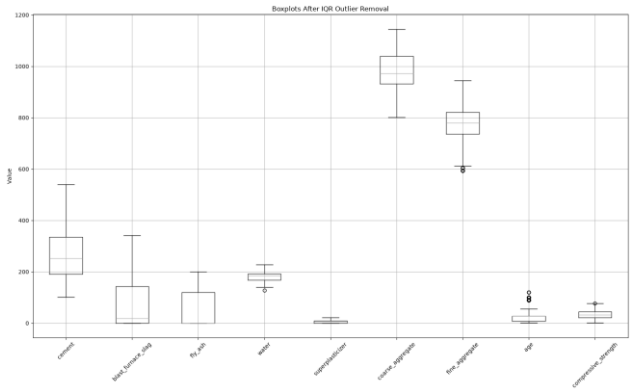
- **Feature–Target Relationships:** Scatter plots were used to observe how each input feature relates to concrete compressive strength.



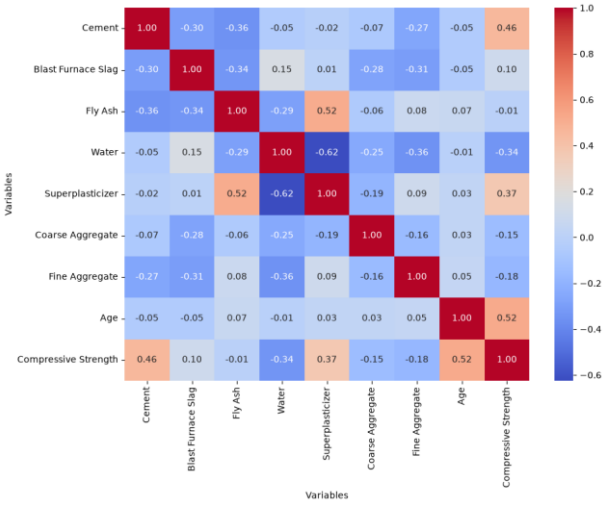
3.2 Data Preprocessing

After completing the exploratory analysis, the dataset was cleaned and prepared for model development using the following steps:

- **Column Standardization:** Column names were cleaned and standardized to ensure consistent naming throughout the project.
- **Data Type Validation:** All expected input features and the target variable were converted and verified as numerical values.
- **Missing Value Handling:** Records were checked for null values, and rows containing missing values would be removed if present.
- **Duplicate Removal:** Duplicate observations were removed to reduce repeated information in the training dataset.
- **Skewness Analysis:** The skewness of each numerical variable was calculated to identify features with asymmetric distributions.
- **Outlier Removal:** The IQR method was applied to detect extreme observations. Rows containing values outside the defined IQR limits were removed.



- **Post-Cleaning Verification:** Boxplots were generated again after outlier removal to inspect the cleaned distributions.
- **Correlation Reassessment:** Correlation analysis was repeated on the cleaned dataset to understand feature relationships after preprocessing.



The cleaned dataset was then saved and used as the input for the feature engineering and model-training stages.

3.3 Feature Engineering

After preprocessing, feature engineering techniques were applied to prepare the input variables for further analysis and machine learning model development. The following techniques were performed:

- **Feature and Target Separation:** The 8 input variables were separated from the target variable, compressive strength, to create the feature matrix X and target vector y.
- **Variance-Based Feature Selection:** Variance Threshold was used to identify and remove features with very low variation, since such features may provide limited information for prediction.
- **Feature Standardization:** StandardScaler was applied to transform the selected features so that they had approximately zero mean and unit variance. This prevents features with larger numerical ranges from dominating scale-sensitive techniques.
- **Dimensionality Reduction using PCA:** Principal Component Analysis (PCA) was applied to transform the selected standardized features into a smaller set of uncorrelated principal components while retaining at least 95% of the total variance in the data.
- **Explained Variance Analysis:** The explained variance ratio and cumulative explained variance were calculated to determine how much information was retained by each principal component.

- **PCA Loading Analysis:** Feature loadings were examined to identify how strongly the original input variables contributed to each principal component.
- **Processed Dataset Creation:** The standardized features were saved, while the PCA-transformed features together with the target variable were saved for further experimentation.

The feature engineering stage provided both the original cleaned feature representation and transformed feature representations, allowing the group to select appropriate inputs according to the requirements of each regression algorithm.

4. Model Design and Implementation

4.1 Data Splitting and Validation

The dataset was divided into:

- **Training set (80%)** – used for model learning.
- **Testing set (20%)** – used for final performance evaluation.
- A fixed random state of 42 was used to ensure reproducibility.
- Cross-validation was applied during model development and hyperparameter tuning to obtain a more reliable estimate of model performance and reduce overfitting.

4.2 Training and Performance Monitoring

- Each group member developed and trained a separate regression model.
- Baseline models were trained before hyperparameter tuning.
- Hyperparameter tuning was performed to improve the performance of each model where required.
- Model performance was evaluated using MAE, MSE, RMSE, and R^2 Score.
- All models were developed using Python, Scikit-learn, Pandas, and NumPy.
- Performance visualizations were generated using Matplotlib and Seaborn.
- The final model comparison was performed using the same evaluation metrics and test dataset.

4.3 Model Selection and Tuning

Member ID	Model	Key Hyperparameters	Best Parameters Used	Before Tuning	After Tuning
IT25101419	XGBoost Regressor		XGBoost tuned parameter	R^2 = 0.9323	R^2 = 0.9326
IT25102211	KNN Regressor	n_neighbors, weights, p	GridSearchCV	R^2: 0.8191	R^2: 0.8293
IT25103164	Random Forest	n_estimators, max_depth, criterion, random state	GridSearchCV	R^2 = 0.9238	R^2 = 0.9419
IT25100456	Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features	RandomizedSearchCV	R^2 = 0.9238	R^2 = 0.9270
IT25102571	SVR (Support Vector Regression)	C, kernel, gamma, epsilon	GridSearchCV	R^2 - 0.7092	R^2 - 0.8703
IT25100040	Linear Regression	fit_intercept	GridSearchCV	R^2 - 0.7711	R^2 - 0.7711

5. Model Design and Implementation

5.1 Evaluation Metrics

To evaluate the regression models, the following standard regression metrics were used:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between the actual and predicted compressive strength values.
- **Mean Squared Error (MSE):** Measures the average squared difference between actual and predicted values, giving more weight to larger errors.
- **Root Mean Squared Error (RMSE):** Represents prediction error in the same unit as compressive strength and penalizes larger errors.
- **R^2 Score:** Measures how well the model explains the variation in concrete compressive strength. A higher R^2 value indicates better predictive performance.

These metrics are appropriate for this project because concrete compressive strength is a continuous numerical target variable.

5.2 Model performance Comparison

Model	↓MAE	↓MSE	↓RMSE	R^2 Score
IT25101419 - XGBoost Regressor	**2.9364**	**17.5896**	**4.1940**	**0.9326**
IT25102211 - KNN	**4.9274**	**44.5186**	**6.6722**	**0.8293**
IT25103164- Random Forest	**2.9273**	**14.3816**	**3.7923**	**0.9419**
IT25100456 - Random Forest	**3.1411**	**19.0406**	**4.3636**	**0.9270**
IT25102571 - SVR	**4.2173**	**33.8302**	**5.8164**	**0.8703**
IT25100040 - Linear Regression	**6.0100**	**59.7150**	**7.7275**	**0.7711**

For comparison, models with lower MAE, MSE, and RMSE and a higher R^2 Score indicate stronger predictive performance

5.1 Evaluation Summary

Based on the final model performance comparison, **Random Forest Regressor** achieved the best overall performance, with the **highest R^2 score of 0.9419** and the lowest error values (**MAE = 2.9273, MSE = 14.3816, RMSE = 3.7923**). This indicates that it provided the most accurate predictions of concrete compressive strength among the six models.
XGBoost Regressor was the second-best model with an **R^2 score of 0.9326**, followed closely by **Random Forest Regressor** with an **R^2 score of 0.9270**. The SVR and KNN models produced moderate performance, while **Linear Regression** showed the weakest performance with the lowest R^2 score of **0.7711** and the highest prediction errors.
Overall, the results show that **ensemble tree-based models, particularly Random Forest and XGBoost, performed better than the other regression algorithms** for predicting concrete compressive strength.
Conclusion

1. **Best Overall Model:** Random Forest Regressor –(**R^2 = 0.9419**)
2. **Strong Alternatives:** XGBoost Regressor (**R^2 = 0.9326**) and Random Forest – (**R^2 = 0.9270**)
3. **Moderate Performers:** SVR and KNN
4. **Lowest Performer:** Linear Regression (**R^2 = 0.7711**)

6. Ethical Considerations and Bias Mitigation

6.1 Dataset Limitations

- The data represents laboratory concrete measurements and may not fully represent every real construction environment.
- Environmental factors such as temperature and humidity are not included in the dataset. Proposal Presentation

6.2 Prediction Errors and Safety

- Machine learning predictions may contain estimation errors.
- The predicted strength should be used as a decision-support estimate, not as a replacement for certified laboratory testing.
- Final engineering decisions should remain under professional supervision.

6.3 Bias Mitigation

- Multiple regression models were trained and compared.
- The same evaluation metrics were used for fair comparison.
- Cross-validation and hyperparameter tuning were used to reduce overfitting.

6.4 Transparency

- Model performance was reported using MAE, MSE, RMSE, and R^2 Score.
- Prediction errors and model limitations were clearly considered when interpreting the results.

6.5 Ethical Summary

The project was conducted with attention to responsible and transparent use of machine learning. Although the assigned dataset provides useful information for predicting concrete compressive strength, the model may not capture every real-world condition. Therefore, the predictions should be interpreted carefully and used only as supporting information. Model limitations, evaluation results, and possible prediction errors should be clearly communicated, while final engineering decisions should continue to rely on appropriate professional judgment and testing.

7. AI Tool Usage Declaration and Transparency

Generative AI tools were used as supporting tools during the project for coding, debugging, explanations, model development, and documentation. All AI-assisted outputs were reviewed, tested, and verified by the group before being included in the final project.

AI Tool	Version	Supported Activities	Extent of Use	Verification and Group Ownership
ChatGPT	GPT-5.6 Sol	ML concept explanations, Python debugging, EDA and preprocessing guidance, model training support, report writing and editing	Extensive	Generated suggestions were reviewed by group members, code was executed and tested, and final decisions were made by the group
GitHub Copilot		notebook assistance, code suggestions, and workflow support	Moderate	Suggested code was reviewed and tested before use
Google Gemini /Colab AI	Google Gemini integrated with Google Colab	Notebook support, code suggestions, and troubleshooting	Minimal	Suggestions were tested in the notebook

Transparency Statement:

AI tools were used only as development assistants. The group maintained responsibility for understanding the code, running experiments, validating results, correcting errors, interpreting model performance, and making final project decisions. No AI-generated output was included without review and verification by the group.

8. AI Tool Usage Declaration and Transparency

8.1 Team Collaboration and Roles

- The project was completed collaboratively by six group members.
- Data exploration, preprocessing, feature engineering, model training, evaluation, and documentation were shared among the team.
- Each member was responsible for training and evaluating a different regression model.
- Model results were combined to identify the best-performing approach

8.2 Challenges and Problem Solving

- Data Quality: Duplicate records and outliers had to be identified and handled before model training.
- Feature Engineering: Scaling, variance-based feature selection, and PCA were explored to prepare the data.
- Model Performance: Hyperparameter tuning and cross-validation were used to improve model performance and reduce overfitting.
- Fair Model Comparison: The same train-test split and common evaluation metrics were used across models.

8.3 Lessons Learned

- Gained practical experience in EDA, preprocessing, feature engineering, and regression modeling.
- Learned the importance of selecting suitable preprocessing techniques for different algorithms.
- Improved understanding of MAE, MSE, RMSE, R^2 Score, cross-validation, and hyperparameter tuning.
- Developed practical skills in Python, Scikit-learn, Pandas, NumPy, Matplotlib, Git, and GitHub.
- Improved teamwork, communication, and problem-solving skills.

8.4 Future Improvements

- Train the models using a larger and more diverse concrete dataset.
- Include additional factors such as temperature, humidity, and curing conditions.
- Explore more advanced regression algorithms and tuning techniques.
- Improve model explainability and develop a user-friendly prediction interface.
- Validate the final model using real-world construction data before practical deployment.

9. References

1. **Yeh, I. C. (1998). *Modeling of strength of high-performance concrete using artificial neural networks*. Cement and Concrete Research, 28(12), 1797–1808. https://doi.org/10.1016/S0008-8846(98)00165-3 Scribd**
2. **Kaggle. (n.d.). *Concrete Compressive Strength Data Set*. The dataset contains 1,030 observations, eight quantitative input variables, and one compressive-strength output variable. Kaggle**
3. **Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830. Journal of Machine Learning Research**
4. **Matplotlib Development Team. (n.d.). *Matplotlib documentation*. Used for data visualization and model-performance graphs.**
5. **pandas development team. (n.d.). *pandas documentation*. Used for dataset loading, cleaning, preprocessing, and data manipulation.**
6. **NumPy Developers. (n.d.). *NumPy documentation*. Used for numerical computations during preprocessing and model evaluation.**